EEE/CPE 64 — Lab Report 6

1-bit Comparator

Student: Yeshua Macwan
Instructor: Riaz Ahmad

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Introduction

Introduction

In this lab I will design, simulate, and implement a 1-bit comparator. First, I'll derive the truth table and Boolean equations and sketch the gate-level schematic. Then I'll build and test the circuit in Multisim (using switches and LEDs), replicate it on a breadboard, and finally write Verilog in Quartus to program an FPGA, mapping switches to inputs and LEDs to outputs. I'll verify all four input cases (00, 01, 10, 11) at each stage and document results with photos.

Preliminary Calculations

2.1 Gate-level Schematics

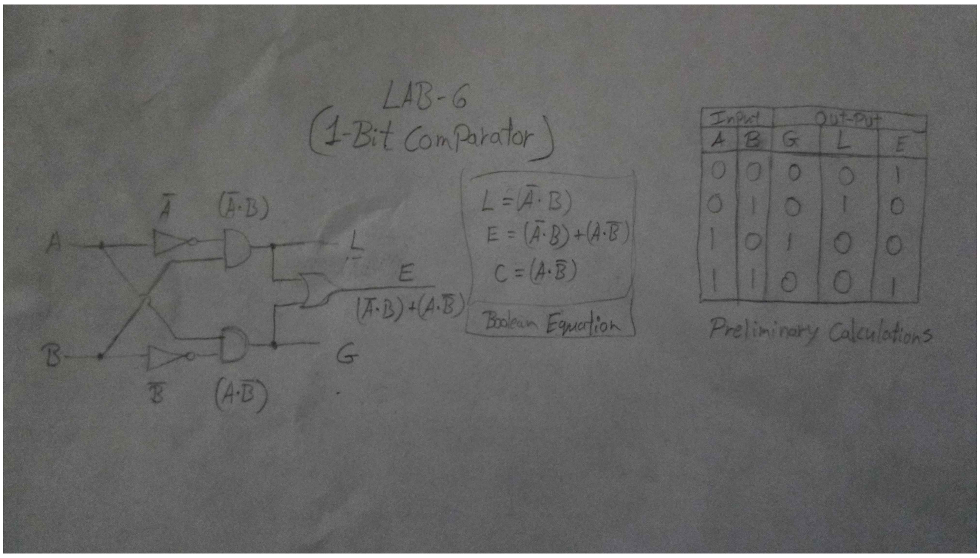
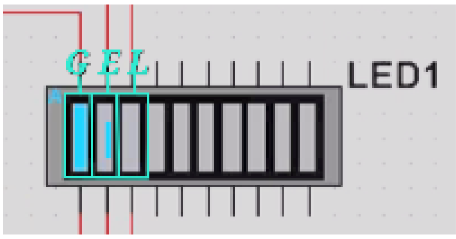


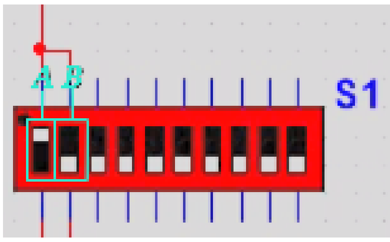
Figure 1: Gate level Schematics

A 1-bit comparator takes inputs A and B. Inverters produce $\bar{A}$ and $\bar{B}$; two AND gates give $G = A\bar{B}$ (meaning $A > B$) and $L = \bar{A}B$ (meaning $A < B$). The third output is $E = G + L = \bar{A}B + A\bar{B} = A \oplus B$, so it is high only when the inputs differ.

Simulation Results

3.1 NI Multisim: Schematics





The first switch is A and the second switch is B. NOT gates create $\bar{A}$ and $\bar{B}$. An AND gate with A and $\bar{B}$ gives G (interpreted as $A > B$), another AND with $\bar{A}$ and B gives L ($A < B$), and a NOR of G and L produces E ($A = B$). The LEDs display G, L, and E.

3.2 Simulation Data

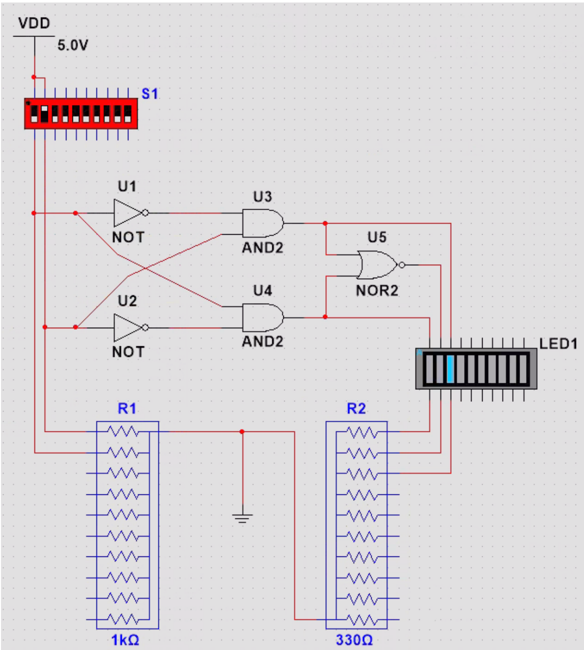


Image 5

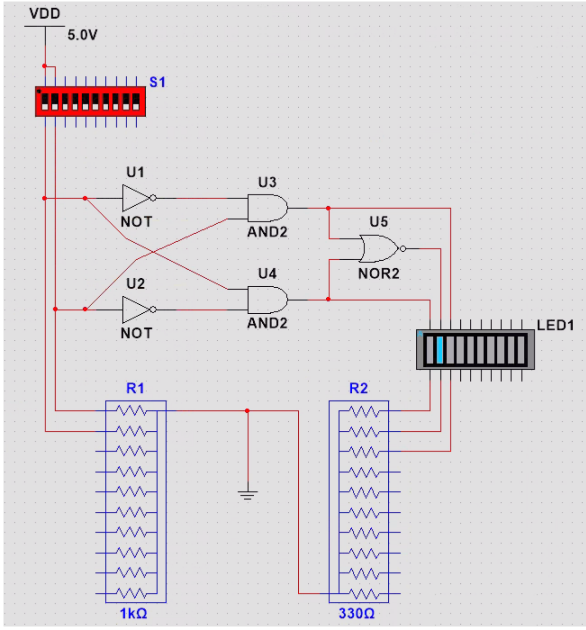


Image 6

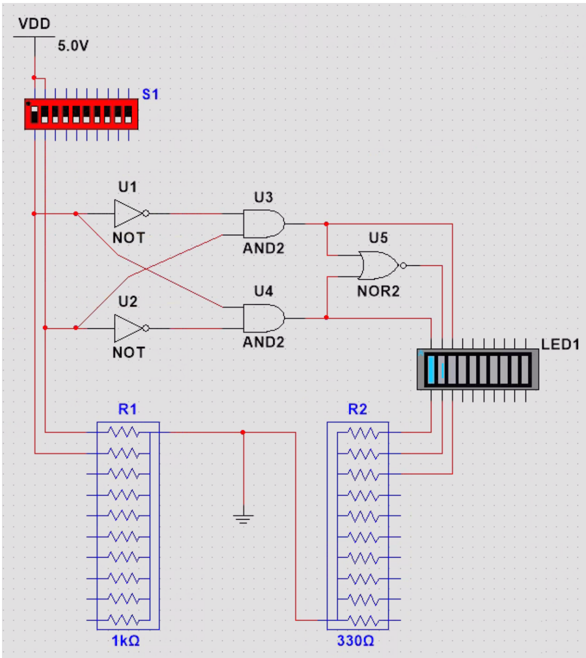


Image 7

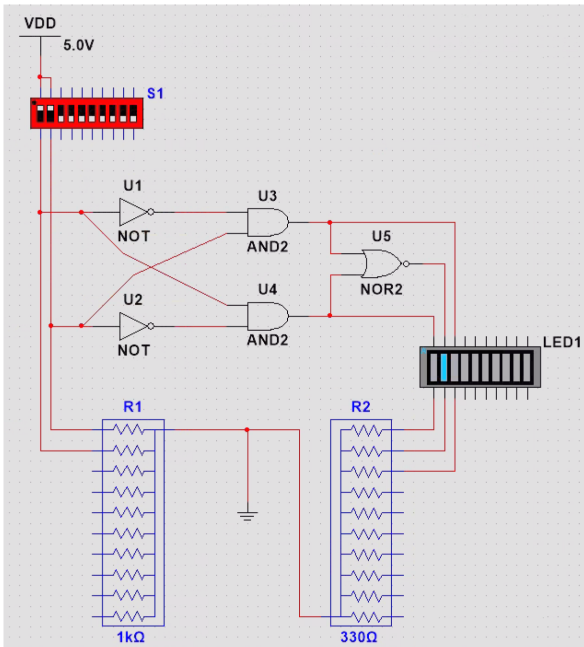
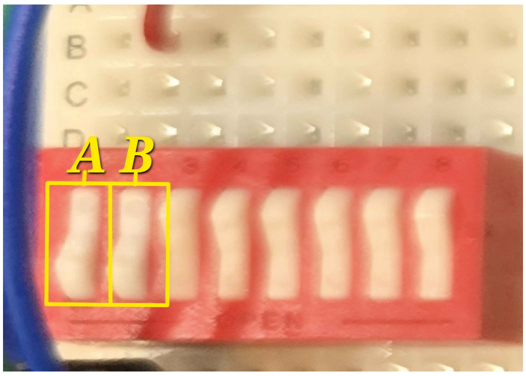


Image 8

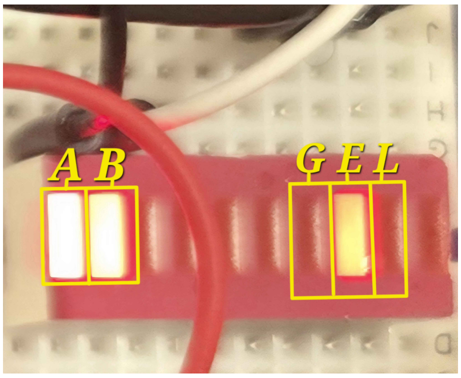
Figure 3: 1-bit Comparator

Bread Board Results

4.1 Bread board Results



(a) Image 1



(b) Image 2

Figure 4: Comparison of two simulation results.

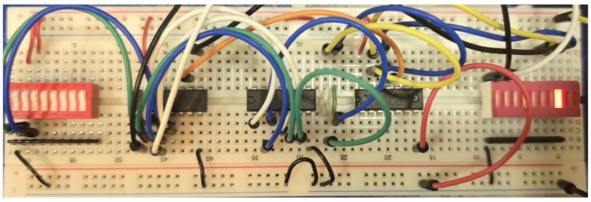


Image 5

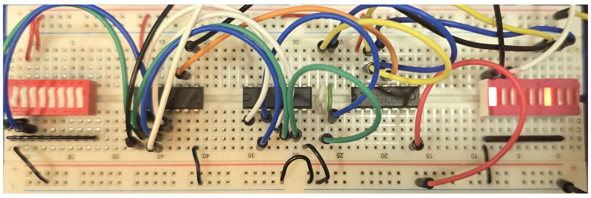


Image 6

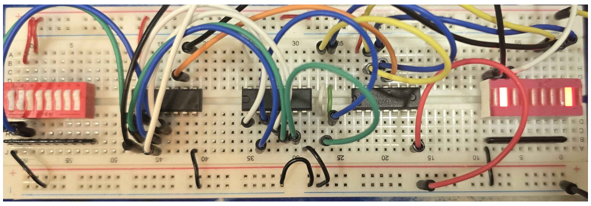


Image 7

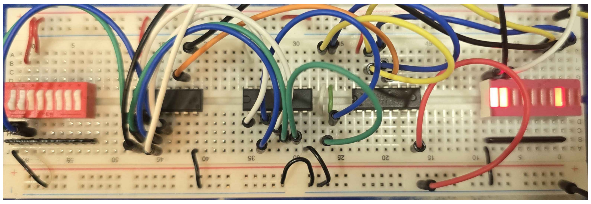


Image 8

Figure 5: 1-bit Comparator

FPGA RESULTS

5.1 FPAG RESULTS

Forgot to take pictures of it.

CONCLUSION

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We validated the 1-bit comparator in three stages: preliminary calculations, NI Multisim simulation, and a breadboard build. For all four input pairs (00, 01, 10, 11), the observed behavior matched the predicted truth table—$A > B$ only for 10, $A < B$ only for 01, and equality for 00 and 11. The agreement across analysis, simulation, and hardware confirms the correctness of the derived logic and its implementation.

A	B	Preliminary Analysis	Simulation Results	Breadboard
0	0	$A = B$	$A = B$	$A = B$
0	1	$A < B$	$A < B$	$A < B$
1	0	$A > B$	$A > B$	$A > B$
1	1	$A = B$	$A = B$	$A = B$

Table 1: 1-bit comparator results confirmed across preliminary work, Multisim, and breadboard.